

10-WATT AUDIO AMPLIFIER Using LM1875

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Here is a simple 10-watt audio amplifier using IC LM1875. The IC has five pins and can deliver a maximum of 18-watt output power depending upon the supply voltage and load. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the audio amplifier is shown in Fig. 2. It is built around popular amplifier LM1875

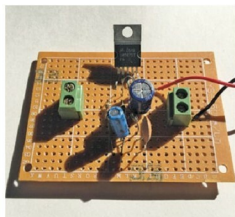


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

IC1 - LM1875 AF amplifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R2 - 82-kilo-ohm

R3 - 4.7-kilo-ohm

R4 - 2.7-ohm

Capacitors:

C1 - 4.7 μ F, 25V electrolytic

C2 - 220 μ F, 35V electrolytic

C3 - 1000 μ F, 35V electrolytic

C4 - 0.22 μ F ceramic disk

Miscellaneous:

CON1 - 2-pin connector

LS1 - 8-ohm, 10-watt

S1 - On/off switch

BATT.1 - 15V battery/15V DC adaptor

2-pin terminal connector for battery

Heat-sink for LM1875

Audio signal from audio sources like mobile phone, laptop or MP3 player

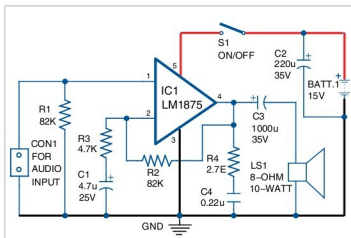


Fig. 2: Circuit diagram of the simple 10-watt audio amplifier using LM1875

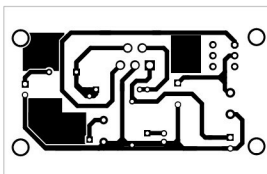


Fig. 3: PCB layout of the audio amplifier

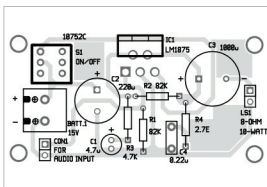


Fig. 4: Components layout for the PCB

(IC1), an 8-ohm, 10-watt speaker (LS1) and a few other components.

Connect the audio input from your mobile/laptop/PC to CON1. Connect 15V battery and switch on

S1. You should hear amplified sound in speaker LS1.

Construction and testing

Use a 25-watt soldering iron for soldering LM1875 and other components. A 2-pin connector should be used for input and output connections to make the prototype safe and clean. Use a suitable heat-sink

for LM1875 to prevent its overheating.

An actual-size, PCB layout for the amplifier is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box. Fix LS1 on the rear side of the box.

For testing the circuit, connect a 6V battery to this circuit along with an 8-ohm, 10-watt speaker.

Take a metal screwdriver and gently touch at the input terminal. If your prototype is working, you will hear a humming sound from the speaker. You may connect a 100-kilo-ohm potmeter between audio input and ground and middle pin of potmeter to pin 1 for volume control. Your circuit is now ready to use.

You can use the power supply from a 5V to 15V battery, which will give you clear and noise-free audio signal output. **EFY**



Raj K. Gorkhali is a regular contributor to EFY. His interests include designing electronics circuits